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CAN PRESS

Abstract

Disclosed is a can press, which comprises a supporting part, a power part and a turnover part. The supporting part comprises a placing plate and a supporting piece, and the placing plate is movably connected with the supporting piece; the power part comprises an extruding piece and a driving piece, the extruding piece is slidably connected to the supporting piece, the driving piece is used for driving the extruding piece to slide, the extruding piece is used for extruding a pop can, after the extrusion, the extruding piece slides to drive the turnover part to move, and the turnover part drives the placing plate to turn over, so that the pop can falls off from the placing plate. The mode is convenient for a user to apply force, saves physical strength, and reduces fatigue of arms of the user.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of Chinese Patent Application No. 202410241707.5 filed on Mar. 1, 2024, the contents of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present invention relates to the technical field of articles of daily use, and more particularly, to a can press.

BACKGROUND

[0003] Pop cans and plastic bottles are the most commonly used food containers in daily life. After people use the pop cans and the bottles, as bodies of the pop cans and the bottles are both cylindrical, they are not easy to place and roll, which are extremely inconvenient to store. Therefore, there is a machine for extruding pop cans into a flat shape in the market, that is, a can press. The can press in the prior art usually uses an extruding piece to extrude a pop can, and a user manually pulls the extruding piece to a certain height, and then slides the extruding piece in a direction close to the pop can to extrude the pop can, and a pressed product cannot be automatically collected and need to be taken manually. When using this method, it is necessary to manually pull up the extruding piece, and arms of the user move linearly in a height direction, which is very laborious. Moreover, the arms are easy to be tired and sore, and the pop can needs to be taken manually, and cannot be automatically placed in a garbage bin or a collection bag. Moreover, the user needs to apply a force vertically downward, which is very inconvenient to operate, and there is one more action to take the product, resulting in low efficiency.

SUMMARY

[0004] In order to solve at least one problem in the prior art, the present invention provides a can press.

[0005] A can press comprises a supporting part, a power part and a turnover part, wherein the supporting part comprises a placing plate and a supporting piece, the placing plate is movably connected with the supporting piece, and the placing plate is used for placing the pop can; the power part comprises an extruding piece and a driving piece, the extruding piece is slidably connected to the supporting piece, and the driving piece is respectively connected with the extruding piece and the supporting piece; the driving piece is used for driving the extruding piece to slide in a direction close to or away from the placing plate, and the extruding piece is used for extruding the pop can; and the turnover part is respectively connected with the extruding piece and the placing plate; wherein, after the extruding piece performs extrusion, the extruding piece slides in the direction away from the placing plate, the extruding piece drives the turnover part to move, and the turnover part drives the placing plate to turn over, so that the pop can falls off from the placing plate.

[0006] In some embodiments, the driving piece comprises a rotating piece and a first connecting piece, the rotating piece is rotatably connected to the supporting piece, and the first connecting piece is respectively connected with the rotating piece and the extruding piece; wherein, the rotating piece rotates to drive the first connecting piece to move, and the first connecting piece moves to drive the extruding piece to slide, so that the extruding piece is close to or away from the placing plate.

[0007] In some embodiments, the supporting part further comprises a first elastic piece, one end of the placing plate away from the supporting piece is movably connected with the supporting piece, one end of the placing plate close to the supporting piece is overlapped with the supporting piece, and the first elastic piece is connected between the placing plate and the supporting piece, wherein

when the extruding piece slides in the direction away from the placing plate, the extruding piece drives the turnover part to move, the turnover part drives the placing plate to rotate in a direction away from the supporting piece, and the first elastic piece drives the placing plate to rotate in a direction close to the supporting piece.

[0008] In some embodiments, the turnover part comprises a sliding piece, the sliding piece is rotatably connected with the placing plate, the sliding piece is slidably connected to the supporting piece, and the sliding piece and the extruding piece are arranged on two opposite sides of the supporting piece in a lengthwise direction at interval, the supporting piece is provided with a through groove, and at least part of the sliding piece is in sliding fit with the through groove and passes through the through groove for abutting against the extruding piece; [0009] wherein, when the extruding piece slides in the direction close to the placing plate, the extruding piece presses the sliding piece to move the sliding piece in a direction away from the extruding piece; and when the extruding piece slides in the direction away from the placing plate, the clamping piece abuts against the sliding piece and drives the sliding piece to slide in the direction away from the extruding piece, and the sliding piece slides to drive the placing plate to rotate.

[0010] In some embodiments, the turnover part comprises a clamping piece and a clamping portion, the clamping piece is connected with the extruding piece and is slidably connected with the supporting piece, the clamping portion is connected to the placing plate and is slidably connected with the supporting piece, and the clamping piece is used for clamping with the clamping portion; [0011] wherein, when the extruding piece slides in the direction close to the placing plate, the extruding piece drives the clamping piece to slide in the direction close to the placing plate, so that the clamping piece is clamped with the clamping portion; and when the extruding piece slides in the direction away from the placing plate, the clamping piece drives the clamping portion to move, and drives the placing plate to rotate relative to the supporting piece.

[0012] In some embodiments, the clamping piece comprises a body portion and two clamping arms symmetrically connected to the body portion, the body portion is connected with the extruding piece, and the two clamping arms protrude in a direction close to each other to form a convex portion, the convex portion is provided with a first inclined plane, an arc surface and a second inclined plane, the first inclined plane, the arc surface and the second inclined plane are connected smoothly in turn, and the convex portion is used for clamping with the clamping portion.

[0013] In some embodiments, the sliding piece bulges in a direction close to the through groove to form an abutting portion, and the abutting portion is in sliding fit with the through groove and passes through the through groove for abutting against the extruding piece; and along a height direction of the supporting piece, the abutting portion gradually inclines from top to bottom in a direction close to the extruding piece.

[0014] In some embodiments, the supporting part comprises a guide plate, one end of the guide plate is connected with the supporting piece, and along a height direction of the supporting piece, the guide plate gradually inclines from bottom to top in a direction close to the sliding piece; wherein, when the extruding piece drives the sliding piece to move in the direction away from the placing plate, the guide plate abuts against the sliding piece to move the sliding piece in the direction away from the extruding piece.

[0015] In some embodiments, the turnover part comprises a second elastic piece, the second elastic piece is connected with the supporting piece, the sliding piece is arranged between the supporting piece and the second elastic piece, and the second elastic piece is used for elastically abutting against the sliding piece to move the sliding piece in a direction close to the through groove.

[0016] In some embodiments, the turnover part further comprises a second connecting piece, the second connecting piece is rotatably connected with the placing plate and the sliding piece respectively, the second connecting piece is formed with a limiting groove, and at least part of the placing plate is located in the limiting groove and is in limiting fit with the second connecting piece; wherein, when the sliding piece slides in the direction away from the placing plate, the

sliding piece drives the second connecting piece to move, and the second connecting piece drives the placing plate to rotate through the limiting groove.

[0017] Compare with the prior art, the can press according to the embodiments of the present invention has the beneficial effects that: the present application comprises the supporting part, the power part and the turnover part. The supporting part comprises the placing plate and the supporting piece, and the placing plate is movably connected with the supporting piece; the power part comprises the extruding piece and the driving piece, the extruding piece is slidably connected to the supporting piece, the driving piece is respectively connected with the extruding piece and the supporting piece, the driving piece is used for driving the extruding piece to slide in the direction close to or away from the placing plate, the extruding piece is used for extruding the pop can, and the turnover part is respectively connected with the extruding piece and the placing plate. When the extruding piece performs extrusion, the extruding piece slides in the direction away from the placing plate, the extruding piece drives the turnover part to move, and the turnover part drives the placing plate to turn over, so that the pop can falls off from the placing plate. Compared with the prior art, the present application drives the extruding piece to slide by rotating the driving piece, so that the pop can is extruded by the extruding piece, and a user can realize an extruding function of the extruding piece only by rotating the driving piece, which is convenient for the user to apply a force, saves physical strength and reduces fatigue of arms of the user; and after the extrusion, the turnover part is driven to move at the same time during a resetting process of the extruding piece, so that the turnover part drives the placing plate to turn over, and the placing plate directly pushes the extruded finished product into a garbage bag or a garbage bin on the ground, thereby realizing a function of automatic collection and reducing a time for taking the extruded finished product and placing a new product to be extruded, and being more convenient to operate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 is a schematic structural diagram of a can press in the embodiments of the present application;

[0019] FIG. 2 is a schematic diagram showing a working state of the can press in the embodiments of the present application;

[0020] FIG. 3 is a schematic structural diagram of a turnover part in the embodiments of the present application;

[0021] FIG. 4 is a cross-section schematic diagram of the can press in the embodiments of the present application;

[0022] FIG. 5 is an enlarged drawing of a structure at a portion B in FIG. 4;

[0023] FIG. 6 is an enlarged drawing of a structure at a portion A in FIG. 3;

[0024] FIG. 7 is an enlarged drawing of a structure at a portion C in FIG. 4;

[0025] FIG. 8 is another schematic structural diagram of the can press in the embodiments of the present application;

[0026] FIG. 9 is another schematic structural diagram of the turnover part in the embodiments of the present application;

[0027] FIG. 10 is a schematic diagram showing another working state of the can press in the embodiments of the present application;

[0028] FIG. 11 is an enlarged drawing of a structure at a portion D in FIG. 10; and

[0029] FIG. 12 is an enlarged drawing of a structure at a portion E in FIG. 10.

[0030] Reference numerals: 1 refers to supporting part; 2 refers to power part; 3 refers to turnover part; X refers to lengthwise direction; Y refers to width direction; Z refers to height direction; 11 refers to supporting piece; 12 refers to placing plate; 13 refers to guide plate; 14 refers to first

elastic piece; **111** refers to through groove; **112** refers to insertion groove; **114** refers to vertical plate; **115** refers to horizontal plate; **121** refers to inserting portion; **122** refers to free end; **123** refers to connecting end; **21** refers to extruding piece; **22** refers to driving piece; **221** refers to rotating piece; **222** refers to first connecting piece; **31** refers to sliding piece; **32** refers to second elastic piece; **33** refers to second connecting piece; **34** refers to clamping piece; **35** refers to clamping portion; **311** refers to abutting portion; **331** refers to limiting groove; **341** refers to body portion; **342** refers to clamping arm; **343** refers to convex portion; **344** refers to first inclined plane; **345** refers to arc surface; and **346** refers to second inclined plane.

DETAILED DESCRIPTION

[0031] In order to make those skilled in the art better understand the technical solution of the present invention, the present invention is described in detail hereinafter with reference to the drawings, and the descriptions in this part are only exemplary and explanatory, and should not limit the scope of protection of the present invention.

[0032] It should be noted that: similar reference numerals and letters indicate similar items in the following drawings, so once one item is defined in one drawing, it does not need to be further defined and explained in the following drawings.

[0033] It should be noted that if the orientation or position relationship indicated by the terms “center”, “upper”, “lower”, “left”, “right”, “vertical”, “horizontal”, “inner”, “outer”, and the like is based on the orientation or position relationship shown in the accompanying drawings, or the orientation or position relationship of the product of the present invention in use, it is only for the convenience of description of the present invention and simplification of the description, and it is not to indicate or imply that the indicated device or element must have a specific orientation, and be constructed and operated in a specific orientation. Therefore, the terms should not be understood as limiting the present invention. In addition, the terms “first”, “second”, “third”, and the like are used for distinguishing description only and cannot be understood as indicating or implying relative importance.

[0034] In addition, the terms “horizontal”, “vertical”, “overhanging”, and the like do not mean that the components are required to be absolutely horizontal or overhanging, but may be slightly inclined. For example, “horizontal” only means that the direction is more horizontal relative to “vertical”, and does not mean that the structure must be completely horizontal, but may be slightly inclined.

[0035] Unless otherwise defined, all technical and scientific terms used herein have the same meanings as those commonly understood by those skilled in the technical field of the present invention. The terms used in the specification of the present invention herein are only for the purpose of describing specific embodiments, and are not intended to limit the present invention.

[0036] The present invention will be further described hereinafter in detail with reference to the drawings.

[0037] Referring to FIG. 1, for the sake of clear description, X indicates a lengthwise direction X of a supporting piece **11**, Y indicates a width direction Y of the supporting piece **11**, and Z indicates a height direction Z of the supporting piece **11**.

[0038] Referring to FIG. 1 and FIG. 2, a can press comprises a supporting part **1**, a power part **2** and a turnover part **3**. The supporting part **1** comprises a placing plate **12** and a supporting piece **11**. The placing plate **12** is movably connected with the supporting piece **11** and is used for placing the pop can. The power part **2** comprises an extruding piece **21** and a driving piece **22**, the extruding piece **21** is slidably connected to the supporting piece **11**, and the driving piece **22** is respectively connected with the extruding piece **21** and the supporting piece **11**, wherein the driving piece **22** is used for driving the extruding piece **21** to slide in a direction close to or away from the placing plate **12**, and the extruding piece **21** is used for extruding the pop can. The turnover part **3** is respectively connected with the extruding piece **21** and the placing plate **12**; wherein, after the extruding piece **21** performs extrusion, the extruding piece **21** slides in the direction away from the

placing plate **12**, the extruding piece **21** drives the turnover part **3** to move, and the turnover part **3** drives the placing plate **12** to turn over, so that the pop can falls off from the placing plate **12**.
[0039] During actual application, the supporting piece **11** is provided with a plurality of fixing holes, and the supporting piece **11** can be fixed to a wall through the fixing holes. The placing plate **12** is placed on the supporting piece **11** and movably connected with the supporting piece **11**. The movable connection particularly refers to that the placing plate **12** can move or rotate relative to the supporting piece **11**, and the placing plate **12** is used for placing the pop can or a plastic bottle. As a main structural part, the supporting piece **11** is used to be fixed to the wall to complete installation of the can press on one hand, and meanwhile, a use space of the can press is saved by installing the supporting piece on the wall. On the other hand, the supporting piece **11** is used for supporting the placing plate **12** to improve a stress capacity of the placing plate **12**.

[0040] In one embodiment, the supporting piece **11** may comprise a horizontal plate **115** and a vertical plate **114** which are vertically connected together. The horizontal plate **115** is movably connected with the placing plate **12** and used for supporting the placing plate **12** to improve the stress capacity of the placing plate **12**. The extruding piece **21** is slidably connected with the vertical plate **114**, the vertical plate **114** is connected with the wall, and the vertical plate **114** may be designed to be parallel to the wall, which can reduce an overall thickness of the can press and save a production cost.

[0041] The above-mentioned extruding piece **21** can slide along the height direction Z of the supporting piece **11**, and the driving piece **22** is respectively connected with the extruding piece **21** and the supporting piece **11**. Specifically, the driving piece **22** may be rotatably connected with the extruding piece **21** and the supporting piece **11**, and the driving piece **22** can rotate in a direction away from the supporting piece **11**, which can drive the extruding piece **21** to slide in a direction close to the placing plate **12**, so that the extruding piece **21** can be extruded into a flat shape for convenient storage. The driving piece **22** rotates in a direction close to the supporting piece **11**, which can drive the extruding piece **21** to slide in a direction away from the placing plate **12**, so that the extruding piece **21** is away from the placing plate **12**, which is convenient for placing next pop can. At the same time, when the extruding piece **21** slides away from the placing plate **12**, the extruding piece **21** drives the turnover part **3** to move. Because the turnover part **3** is connected with the placing plate **12**, the turnover part **3** drives the placing plate **12** to turn over around the horizontal plate **115**, so that an extruded finished product (pop can) can freely fall off from the placing plate **12**, and a function of automatic collection is realized. Compared with the prior art, the present application drives the extruding piece **21** to slide by rotating the driving piece **22**, so that the pop can is extruded by the extruding piece **22**, and a user can realize an extruding function of the extruding piece **21** only by rotating the driving piece **22**, which is convenient for the user to apply a force, saves physical strength and reduces fatigue of arms of the user; and after the extrusion, the turnover part **3** is driven to move at the same time during a lifting process of the extruding piece **21**, so that the turnover part **3** drives the placing plate **12** to turn over, and the placing plate **12** directly pushes the extruded finished product into a garbage bag or a garbage bin on the ground, thereby realizing a function of automatic collection and reducing a time for taking the extruded finished product and placing a non-extruded product, and being more convenient to operate.

[0042] Referring to FIG. 2 and FIG. 3, the driving piece **22** comprises a rotating piece **221** and a first connecting piece **222**. The rotating piece **221** is rotatably connected to the supporting piece **11**, and the first connecting piece **222** is rotatably connected to the rotating piece **221** and the extruding piece **21** respectively; wherein, the rotating piece **221** rotates to drive the first connecting piece **222** to move, and the first connecting piece **222** moves to drive the extruding piece **21** to slide, so that the extruding piece **21** is close to or away from the placing plate **12**.

[0043] During actual application, a bottom of the rotating piece **221** may be rotatably connected to the supporting piece **11**, a middle of the rotating piece **221** may be rotatably connected with a

bottom of the first connecting piece 222, and a top of the first connecting piece 222 is rotatably connected with the extruding piece 21. When the rotating piece 221 rotates in the direction away from the supporting piece 11, the rotating piece 221 drives the bottom of the first connecting piece 222 to move in the direction away from the supporting piece 11, so that the bottom of the first connecting piece 222 generates a downward pulling force on a top thereof. As the top of the first connecting piece 222 is rotatably connected with the extruding piece 21, the first connecting piece 222 generates a downward pulling force on the extruding piece 21, which makes the extruding piece 21 move downwards, that is, move in the direction close to the placing plate 12 to realize an extruding effect of the extruding piece 21.

[0044] When the rotating piece 221 rotates in the direction close to the supporting piece 11, the rotating piece 221 drives the bottom of the first connecting piece 222 to move in the direction close to the supporting piece 11. In this case, the bottom of the first connecting piece 222 generates an upward pushing force on the top thereof. Therefore, the first connecting piece 222 generates an upward pushing force on the extruding piece 21, which makes the extruding piece 21 move upwards, that is, move in the direction away from the placing plate 12 to place next pop can on the placing plate 12. In the present application, the rotating piece 221 drives the extruding piece 21 to slide through the first connecting piece 222, which increases a rotating range of the rotating piece 221. The rotating piece 221 can rotate around the supporting piece 11 by 180 degrees, which increases a movable range of the extruding piece 21. On one hand, a maximum distance between the extruding piece 21 and the placing plate 12 is increased, so that different types of pop cans can be placed. On the other hand, the rotating range of the rotating piece 221 is close to 180 degrees, which increases an extruding force of the extruding piece 21 to extrude the pop can, thus extruding the pop can more tightly and saving a storage space.

[0045] Referring to FIG. 3 and FIG. 4, the supporting part 1 further comprises a first elastic piece 14, and one end of the placing plate 12 away from the supporting piece 11 is movably connected with the supporting piece 11. One end of the placing plate 12 close to the supporting piece 11 is overlapped with the supporting piece 11, and the first elastic piece 14 is connected between the placing plate 12 and the supporting piece, wherein, when the extruding piece slides in the direction away from the placing plate, the extruding piece drives the turnover part to move, the turnover part 3 drives the placing plate 12 to rotate in the direction away from the supporting piece 11, and the first elastic piece 14 drives the placing plate 12 to rotate in the direction close to the supporting piece 11.

[0046] Specifically, one end of the placing plate 12 close to the vertical plate 114 is a free end 122, and the free end 122 is overlapped with the horizontal plate 115. One end of the placing plate 12 away from the vertical plate 114 is a connecting end 123, and the connecting end 123 is rotatably connected with the horizontal plate 115. In this manner, the free end 122 of the placing plate 12 can rotate around the horizontal plate 115. The turnover part 3 is connected with the placing plate 12 and the extruding piece 21 respectively. When the extrusion is completed, the rotating piece 221 drives the extruding piece 21 to slide in the direction away from the placing plate 12, and the extruding piece 21 drives the turnover part 3 to move, so that the turnover part 3 drives the placing plate 12 to turn over around the horizontal plate 115, so that the pop can falls off from the placing plate 12, and a function of automatic collection is realized. Since the first elastic piece 14 is arranged between the placing plate 12 and the supporting piece 11, when the pop can falls off, the placing plate 12 is driven to rotate in the direction close to the supporting piece 11 by elastic resilience of the first elastic piece 14 until the free end 122 is overlapped on the horizontal plate 115 again, thus realizing a function of automatic reset. A spring may be employed as the first elastic piece 14.

[0047] In one embodiment, referring to FIG. 2 and FIG. 3, the turnover part 3 may comprise a sliding piece 31. The sliding piece is rotatably connected with the placing plate 12, and the sliding piece 31 is slidably connected to the supporting piece 11. Moreover, the sliding piece 31 and the

extruding piece **21** are arranged on two opposite sides of the supporting piece **11** in a lengthwise direction X at interval, and the supporting piece **11** is provided with a through groove **111**. At least part of the sliding piece **31** is in sliding fit with the through groove **111** and passes through the through groove **111** for abutting against the extruding piece **21**. It can be understood that the through groove **111** extends along the height direction Z of the supporting piece **11**, and the sliding piece **31** may slide in the through groove **111**.

[0048] When the extruding piece **21** slides in the direction close to the placing plate **12**, the extruding piece **21** presses the sliding piece **31** to move the sliding piece **31** in a direction away from the extruding piece **21**. Specifically, the sliding piece **31** is provided with a top and a bottom, the bottom is slidably connected with the supporting piece **11** and is rotatably connected with the free end **122** of the placing plate **12**. At least part of the top passes through the through groove **111**, and the top may be made of plastic or soft rubber. In an initial state, the top is clamped with a groove bottom of the through groove **111**, that is, the sliding piece **31** can only slide upwards in the height direction Z in the through groove **111**. When the extruding piece **21** slides in the direction close to the placing plate **12**, the extruding piece **21** extrudes the top of the sliding piece **31** downwards. Because the top is the free end **122** and has certain elasticity, the top of the sliding piece **31** can move in the direction away from the extruding piece **21** along the length direction X of the supporting piece **11** to avoid the extruding piece **21** and avoid obstructing the extruding piece **21** from moving downwards. Until the extruding piece **21** moves downwards to separate from the sliding piece **31**, the top of the sliding piece **31** can be reset under the action of the elasticity thereof and extend into the through groove **111** again.

[0049] It can be understood that the top of the sliding piece **31** may be made of elastic materials such as plastic or soft rubber, or the elastic piece can be arranged between the sliding piece **31** and the supporting piece **11**, and the sliding piece **31** can be driven to reset by a restoring force of the elastic piece, so that the sliding piece can extend into the through groove **111** again.

[0050] After the extruding piece **21** extrudes the pop can, the rotating piece **221** drives the extruding piece **21** to slide upwards, that is, the extruding piece **21** slides in the direction away from the placing plate **12**, the extruding piece **21** abuts against the sliding piece **31** and drives the sliding piece **31** to slide in the direction away from the placing plate **12**, and the sliding piece **31** slides to drive the placing plate **12** to rotate.

[0051] Because the top of the sliding piece **31** is clamped with the groove bottom of the through groove **111**, the sliding piece **31** cannot slide downwards, but can slide upwards in the height direction Z in the through groove **111**. Therefore, when the extruding piece **21** slides upwards and abuts against the top of the sliding piece **31**, the extruding piece **21** can continue to slide upwards to drive the sliding piece **31** to slide together. In this manner, the bottom of the sliding piece **31** drives the free end **122** of the placing plate **12** to move upwards, so that the placing plate **12** can be turned over around the horizontal plate **115**, so that the pop can is capable of falling off from the placing plate **12**. By arranging the turnover part **3**, when the extruding piece **21** moves away from the placing plate **12** to prepare for next extruding work after the extruding piece **21** extrudes the pop can, the extruding piece **21** drives the turnover part **3** to turn the placing plate **12**, so that the turnover part **3** drives the placing plate **12** to turn over, and the pop can automatically falls off without manual taking, thus saving time and labor.

[0052] Referring to FIG. 4 and FIG. 5, the sliding piece **31** bulges in a direction close to the through groove **111** to form an abutting portion **311**, and the abutting portion **311** is in sliding fit with the through groove **111** and passes through the through groove **111** for abutting against the extruding piece **21**. Along the height direction Z of the supporting piece **11**, the abutting portion **311** gradually inclines from top to bottom in a direction close to the extruding piece **21**.

[0053] One side of the abutting portion **311** close to the extruding piece **21** is an inclined plane, and the inclined plane gradually inclines to the direction of the extruding piece **21** from top to bottom, so that when the extruding piece **21** moves downwards, the extruding piece **21** abuts against the

inclined plane. Under the guidance of the inclined plane, the extruding piece **21** slides along the inclined plane and exerts an outward abutting force on the inclined plane. Under the action of the abutting force, the abutting portion **311** moves outwards, that is, moves in the direction away from the extruding piece **21**, which can effectively prevent the abutting portion **311** from blocking the movement of the extruding piece **21**, so that the extruding piece **21** slides downwards more smoothly. Meanwhile, as a length of the through groove **111** protruding from a lower side of the abutting portion **311** is longer, when the extruding piece **21** moves upwards, the extruding piece **21** and the abutting portion **311** are more easily abutted together, which effectively prevents the extruding piece **21** from being separated from the abutting portion **311** during the upward movement, so that the extruding piece **21** can drive the abutting portion **311** to slide upwards in the height direction Z, and then drive the placing plate **12** to turn over, thus completing the function of automatic collection.

[0054] Referring to FIG. 6, the supporting part **1** further comprises a guide plate **13**. One end of the guide plate **13** is connected with the supporting piece **11**, and along the height direction Z of the supporting piece **11**, the guide plate **13** gradually inclines from bottom to top in the direction close to the sliding piece **31**. When the extruding piece **21** drives the sliding piece **31** to move in the direction away from the placing plate **12**, the guide plate **13** abuts against the sliding piece **31** to move the sliding piece **31** in the direction away from the extruding piece **21**.

[0055] In practical application, the supporting piece **11** may be provided with a U-shaped through groove **111**, and the U-shaped through groove **111** is arranged around the guide plate **13**. The abutting portion **311** of the sliding piece **31** may be in sliding fit with the through groove **111** on one side of the guide plate **13**. When the extruding piece **21** slides upwards and abuts against the abutting portion **311**, the extruding piece **21** drives the sliding piece **31** to slide upwards. Since the guide plate **13** gradually inclines to an outside of the supporting piece **11** from bottom to top, the sliding piece **31** abuts against the guide plate **13** in an upward sliding process, so that the sliding piece **31** moves in the direction away from the extruding piece **21**, so that the extruding piece **21** can be smoothly separated from the abutting portion **311**. In this way, under the guiding function of the guide plate **13**, when the sliding piece **31** slides upwards to drive the placing plate **12** to turn over, the extruding piece **21** and the abutting portion **311** are separated smoothly, and the first elastic piece **14** pulls the placing plate **12** to reset, further pulling the sliding piece **31** to slide downwards to the initial position. Meanwhile, the abutting portion **311** may extend into the through groove **111** again under an elastic action of the sliding piece **31** or a restoring force of the elastic piece, thus completing the function of automatic reset.

[0056] It can be understood that the sliding piece **31** may be provided with two abutting portions **311**, which are respectively in sliding fit with the through grooves **111** on both sides of the guide plate **13**, and the guide plate **13** is located between the two abutting portions **311**. Such arrangement can also play the above-mentioned guiding function.

[0057] Referring to FIG. 3 again, the turnover part **3** further comprises a second elastic piece **32**. The second elastic piece **32** is connected with the supporting piece **11**, and the sliding piece **31** is arranged between the supporting piece **11** and the second elastic piece **32**. The second elastic piece **32** is used for elastically abutting against the sliding piece **31**, so that the sliding piece **31** moves in the direction close to the through groove **111**.

[0058] In order to make the sliding piece **31** be reset smoothly, a second elastic piece **32** is provided in the present application. The second elastic piece **32** abuts the sliding piece **31** on a side of the supporting piece **11**. When the sliding piece **31** moves to the outside of the supporting piece **11** under the action of the guide plate **13** or the extruding piece **21**, the sliding piece **31** is pushed to move in the direction close to the through groove **111** by elastic resilience of the second elastic piece **32**, so that the sliding piece **31** can be reset. The abutting portion **311** may extend into the through groove **111** again to abut against and cooperate with the extruding piece **21**, so as to prevent the sliding piece **31** from being elastically restored after deformation, and improve a

service life of the can press. A spring leaf may be employed as the second elastic piece 32.

[0059] Referring to FIG. 7, the turnover part 3 further comprises a second connecting piece 33. The second connecting piece 33 is rotatably connected with the placing plate 12 and the sliding piece 31 respectively. The second connecting piece 33 is formed with a limiting groove 331, and at least part of the placing plate 12 is located in the limiting groove 331 and is in limiting fit with the second connecting piece 33. When the sliding piece 31 slides in the direction away from the placing plate 12, the sliding piece 31 drives the second connecting piece 33 to move, and the second connecting piece 33 drives the placing plate 12 to rotate through the limiting groove 331.

[0060] During actual application, when the extruding piece 21 drives the sliding piece 31 to slide upwards along the height direction Z, the sliding piece 31 drives the second connecting piece 33 to move upwards. Because two ends of the second connecting piece 33 are rotatably connected with the sliding piece 31 and the placing plate 12 respectively, and the placing plate 12 is in limiting fit with the limiting groove 331, the second connecting piece 33 moves upwards and abuts against the placing plate 12, driving the free end 122 of the placing plate 12 to turn around the horizontal plate 115, so that the pop can falls off from the placing plate 12. Under the condition that a sliding distance of the sliding piece 31 in the height direction Z is limited, by setting the second connecting piece 33, a turnover angle of the placing plate 12 is increased, which is beneficial for the pop can to fall off.

[0061] As an alternative embodiment, it is different from the above-mentioned embodiment in that: the structure of the turnover part 3 is different. Referring to FIG. 8 and FIG. 9, the turnover part 3 may also comprise a clamping piece 34 and a clamping portion 35. The clamping piece 34 is connected with the extruding piece 21 and is slidably connected with the supporting piece 11. The clamping portion 35 is connected to the placing plate 12 and is slidably connected with the supporting piece 11. The clamping piece 34 is used for clamping with the clamping portion 35.

[0062] During actual application, a position of the clamping piece 34 is opposite to that of the clamping portion 35. When the pop can needs to be extruded, the driving piece 22 drives the extruding piece 21 to slide in the direction close to the placing plate 12, and the extruding piece 21 drives the clamping piece 34 to slide in the direction close to the placing plate 12, so that the clamping piece 34 is clamped with the clamping portion 35. After the extrusion is completed, when the driving piece 22 drives the extruding piece 21 to slide in the direction away from the placing plate 12, the clamping piece 34 clamps the clamping portion 35. Therefore, the clamping piece 34 drives the clamping portion 35 to move. Because the clamping portion 35 is connected to the placing plate 12, the clamping piece 34 drives the placing plate 12 to turn over around the horizontal plate 115 in this manner, so that the pop can falls off.

[0063] Referring to FIG. 10 and FIG. 11, the clamping piece 34 comprises a body portion 341 and two clamping arms 342 symmetrically connected to the body portion 341. The body portion 341 is connected with the extruding piece 21, and the two clamping arms 342 protrude in a direction close to each other to form a convex portion 343. The convex portion 343 is provided with a first inclined plane 344, an arc surface 345 and a second inclined plane 346. The first inclined plane 344, the arc surface 345 and the second inclined plane 346 are connected smoothly in turn, and the convex portion 343 is used for clamping with the clamping portion 35.

[0064] Specifically, the two clamping arms 342 protrude in the direction close to each other to form the convex portion 343, and the two opposite convex portions 343 form a limiting space for limiting the clamping portion 35 from falling out and preventing the clamping portion 35 from being separated from the clamping piece 34. Meanwhile, the convex portion 343 is provided with the first inclined plane 344, the arc surface 345 and the second inclined plane 346 which are smoothly and transiently connected, and the second inclined plane 346 inclines from bottom to top in the direction close to the arc surface 345. In this manner, when the clamping piece 34 moves downwards and contacts with the clamping portion 35, the second inclined plane 346 is beneficial for the clamping portion 35 to be clamped into the clamping piece 34, and reduces a frictional force

between the clamping piece **34** and the clamping portion **35**. The first inclined plane **344** inclines from top to bottom in the direction close to the arc surface **345**. After the placing plate **12** is turned over, the separation of the clamping portion **35** from the clamping piece **34** is facilitated by a guiding function of the first inclined plane **344**.

[0065] In order to separate the clamping portion **35** from the clamping part **34**, the supporting piece **11** is provided with a sliding chute with a certain length, and the clamping part **35** is slidably connected with the sliding chute. The clamping piece **34** drives the clamping portion **35** to slide against a wall of the sliding chute, so that the clamping portion **35** is separated from the clamping piece **34** through an abutting action of the wall of the sliding chute, and the separated clamping portion **35** is reset with the placing plate **12** under an elastic force of the first elastic piece **14**.

[0066] Referring to FIG. **12**, in one embodiment, one end of the placing plate **12** is provided with an inserting portion **121**, and the horizontal plate **115** is provided with an insertion groove **112**. One end of the placing plate **12** is inserted into the insertion groove **112**, and the other end of the placing plate is overlapped on the horizontal plate **115**. When the clamping piece **34** drives the clamping portion **35** to slide upwards, the clamping portion **35** drives the placing plate **12** to turn over around the horizontal plate **115** until the placing plate **12** turns over to that the inserting portion **121** abuts against a groove wall of the insertion groove **112**. Due to a limiting effect of the inserting portion **121** and the insertion groove **112**, the placing plate **12** cannot continue to rotate. In this case, the clamping portion **35** cannot continue to slide upwards, so that the clamping piece **34** can be separated from the clamping portion **35**. By inserting one end of the placing plate **12** into the horizontal plate **115** and overlapping the other end of the placing plate with the horizontal plate **155**, it is beneficial for the placing plate **12** to be reset under the elastic force of the first elastic piece **14**.

[0067] Different from the prior art, the present application provides a can press which comprises a supporting part **1** and a power part **2**. The supporting part **1** comprises a placing plate **12** and a supporting piece **11**. The placing plate **12** is movably connected with the supporting piece **11** and is used for placing the pop can. The power part **2** comprises an extruding piece **21** and a driving piece **22**, the extruding piece **21** is slidably connected to the supporting piece **11**, and the driving piece **22** is rotatably connected with the extruding piece **21** and the supporting piece **11** respectively, wherein the driving piece **22** rotates for driving the extruding piece **21** to slide in a direction close to or away from the placing plate **12**, and the extruding piece **21** is used for extruding the pop can. Compared with the prior art, in the present application, the extruding piece **21** is driven to slide by rotating the driving piece **22**, so that the extruding piece extrudes the pop can, and the user can realize an extruding function of the extruding piece **21** only by rotating the driving piece **22**, which is convenient for the user to apply force, saves physical strength, reduces fatigue of arms of the user, and is convenient to operate.

[0068] In the description of the present invention, it should also be noted that the terms “arrangement”, “installation”, “connected” and “connection” should be understood in a broad sense unless otherwise clearly specified and defined. For example, they may be fixed connection, removable connection or integrated connection; may be mechanical connection or electrical connection; and may be direct connection indirect connection through an intermediate medium, and connection inside two elements. The specific meanings of the above terms in the present invention can be understood in a specific case by those of ordinary skills in the art.

[0069] It should be noted that relational terms herein, such as first, second, and the like, are used merely to distinguish one entity or operation from another entity or operation, and do not necessarily require or imply that there is any such relationship or order between these entities or operations. Furthermore, the terms “comprise”, “include”, or any other variation thereof, are intended to cover a non-exclusive inclusion, so that a process, a method, an article, or equipment that comprises a list of elements not only includes those elements but also includes other elements not expressly listed, or further includes elements inherent to such process, method, article, or

equipment. In a case without further limitations, an element defined by the phrase “comprising one . . .” does not preclude the presence of additional identical elements in the process, method, article, or equipment that includes the element.

[0070] Although the embodiments of the present invention have been shown and described, it is understood by those of ordinary skills in the art that various changes, modifications, substitutions and variations may be made to these embodiments without departing from the principle and spirit of the present invention, and the scope of the present invention is defined by the appended claims and their equivalents.

Claims

1. A can press for extruding a pop can, comprising: a supporting part (1), comprising a placing plate (12) and a supporting piece (11), wherein the placing plate (12) is movably connected with the supporting piece (11), and the placing plate (12) is used for placing the pop can; a power part (2), comprising an extruding piece (21) and a driving piece (22), wherein the extruding piece (21) is slidably connected to the supporting piece (11), and the driving piece (22) is respectively connected with the extruding piece (21) and the supporting piece (11); wherein, the driving piece (22) is used for driving the extruding piece (21) to slide in a direction close to or away from the placing plate (12), and the extruding piece (21) is used for extruding the pop can; and a turnover part (3) respectively connected with the extruding piece (21) and the placing plate (12); wherein, after the extruding piece (21) performs extrusion, the extruding piece (21) slides in the direction away from the placing plate (12), the extruding piece (21) drives the turnover part (3) to move, and the turnover part (3) drives the placing plate (12) to turn over, so that the pop can falls off from the placing plate (12).
2. The can press according to claim 1, wherein the driving piece (22) comprises a rotating piece (221) and a first connecting piece (222), the rotating piece (221) is rotatably connected to the supporting piece (11), and the first connecting piece (222) is respectively connected with the rotating piece (221) and the extruding piece (21); wherein, the rotating piece (221) rotates to drive the first connecting piece (222) to move, and the first connecting piece (222) moves to drive the extruding piece (21) to slide, so that the extruding piece (21) is close to or away from the placing plate (12).
3. The can press according to claim 1, wherein the supporting part (1) further comprises a first elastic piece (14), and one end of the placing plate (12) away from the supporting piece (11) is movably connected with the supporting piece (11); one end of the placing plate (12) close to the supporting piece (11) is overlapped with the supporting piece (11), and the first elastic piece (14) is connected between the placing plate (12) and the supporting piece (11); wherein, when the extruding piece (21) slides in the direction away from the placing plate (12), the extruding piece (21) drives the turnover part (3) to move, the turnover part (3) drives the placing plate (12) to rotate in a direction away from the supporting piece (11), and the first elastic piece (14) drives the placing plate (12) to rotate in a direction close to the supporting piece (11).
4. The can press according to claim 1, wherein the turnover part (3) comprises a sliding piece (31), the sliding piece (31) is rotatably connected with the placing plate (12), the sliding piece (31) is slidably connected to the supporting piece (11), and the sliding piece (31) and the extruding piece (21) are arranged on two opposite sides of the supporting piece (11) in a lengthwise direction (X) at interval, the supporting piece (11) is provided with a through groove (111), and at least part of the sliding piece (31) is in sliding fit with the through groove (111) and passes through the through groove (111) for abutting against the extruding piece (21); wherein, when the extruding piece (21) slides in the direction close to the placing plate (12), the extruding piece (21) presses the sliding piece (31) to move the sliding piece (31) in a direction away from the extruding piece (21); and when the extruding piece (21) slides in the direction away from the placing plate (12), the

extruding piece (21) abuts against the sliding piece (31) and drives the sliding piece (31) to slide in the direction away from the extruding piece (21), and the sliding piece (31) slides to drive the placing plate (12) to rotate.

5. The can press according to claim 1, wherein the turnover part (3) comprises a clamping piece (34) and a clamping portion (35), the clamping piece (34) is connected with the extruding piece (21) and is slidably connected with the supporting piece (11), the clamping portion (35) is connected to the placing plate (12) and is slidably connected with the supporting piece (11), and the clamping piece (34) is used for clamping with the clamping portion (35); wherein, when the extruding piece (21) slides in the direction close to the placing plate (12), the extruding piece (21) drives the clamping piece (34) to slide in the direction close to the placing plate (12), so that the clamping piece (34) is clamped with the clamping portion (35); and when the extruding piece (21) slides in the direction away from the placing plate (12), the clamping piece (34) drives the clamping portion (35) to move, and drives the placing plate (12) to rotate relative to the supporting piece (11).

6. The can press according to claim 5, wherein the clamping piece (34) comprises a body portion (341) and two clamping arms (342) symmetrically connected to the body portion (341), the body portion (341) is connected with the extruding piece (21), and the two clamping arms (342) protrude in a direction close to each other to form a convex portion (343), the convex portion (343) is provided with a first inclined plane (344), an arc surface (345) and a second inclined plane (346), the first inclined plane (344), the arc surface (345) and the second inclined plane (346) are connected smoothly in turn, and the convex portion (343) is used for clamping with the clamping portion (35).

7. The can press according to claim 4, wherein the sliding piece (31) bulges in a direction close to the through groove (111) to form an abutting portion (311), and the abutting portion (311) is in sliding fit with the through groove (111) and passes through the through groove (111) for abutting against the extruding piece (21); and along a height direction (Z) of the supporting piece (11), the abutting portion (311) gradually inclines from top to bottom in a direction close to the extruding piece (21).

8. The can press according to claim 4, wherein the supporting part (1) comprises a guide plate (13), one end of the guide plate is connected with the supporting piece (11), and along a height direction (Z) of the supporting piece (11), the guide plate (13) gradually inclines from bottom to top in a direction close to the sliding piece (31); wherein, when the extruding piece (21) drives the sliding piece (31) to move in the direction away from the placing plate (12), the guide plate (13) abuts against the sliding piece (31) to move the sliding piece (31) in the direction away from the extruding piece (21).

9. The can press according to claim 4, wherein the turnover part (3) comprises a second elastic piece (32), the second elastic piece (32) is connected with the supporting piece (11), the sliding piece (31) is arranged between the supporting piece (11) and the second elastic piece (32), and the second elastic piece (32) is used for elastically abutting against the sliding piece (31) to move the sliding piece (31) in a direction close to the through groove (111).

10. The can press according to claim 4, wherein the turnover part (3) further comprises a second connecting piece (33), the second connecting piece (33) is rotatably connected with the placing plate (12) and the sliding piece (31) respectively, the second connecting piece (33) is formed with a limiting groove (331), and at least part of the placing plate (12) is located in the limiting groove (331) and is in limiting fit with the second connecting piece (33); wherein, when the sliding piece (31) slides in the direction away from the placing plate (12), the sliding piece (31) drives the second connecting piece (33) to move, and the second connecting piece (33) drives the placing plate (12) to rotate through the limiting groove (331).
